Purdue West Lafayette/Indianapolis
School of Electrical and Computer Engineering
Advanced Topics ECE
ECE (WL) 69500
WL Fall 2026

Course Information

Meeting Information: BHEE 222
Meeting Day(s) and Time(s): MWF / 1030

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Computer Security provides an overview of cybersecurity topics for engineers with a special focus in system design and implementation for computing systems. It covers a wide array of topics relating to this, including security fundamentals (e.g., the CIA triad, the 5-stage pentesting process), as well as cryptography, authentication/authorization, network security, trusted hardware, operating system security, among others. The lectures focus on providing a theoretical perspective of the security auditing process, as well as secure systems design. The course also includes hands-on attack assignments, a defense project component, as well as a midterm/final presentation.

Instructor(s) Contact Information

Name: Santiago Torres Arias
Email: torresar@purdue.edu
Office Phone Number: 7654966610
Student Consultation Hours
Monday, Wednesday 10 AM to 10:30-AM, Friday: 3:30 to 5PM

Additional Information
Please see first deck of slides for other ways to contact me

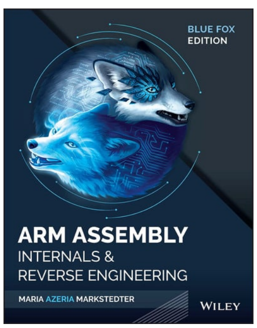
Course Description

Credit Hours: 1.00 to 3.00. Formal classroom or individualized instruction on advanced topics of current interest. Permission of instructor required.

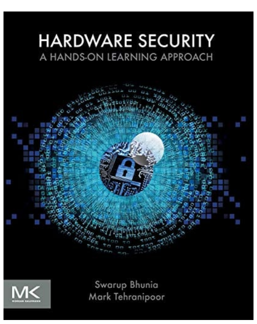
Course Learning Outcomes

By the end of the course, you will be able to:

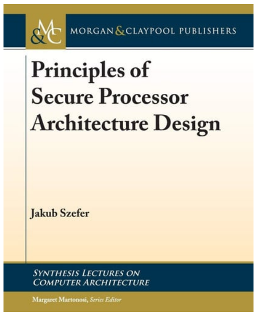
Learning Resources, Technology & Texts



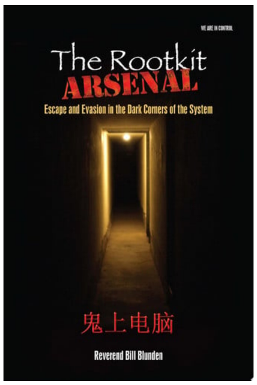
Blue Fox Arm Assembly Internals and Reverse Engineering
ISBN: 9781119745303
Authors: Maria Markstedter
Publisher: Wiley
Publication Date: 2023-04-11
Edition: 1



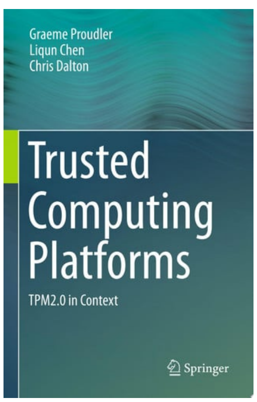
Hardware Security A Hands-on Learning Approach
ISBN: 9780128124772
Authors: Swarup Bhunia, Mark M. Tehranipoor
Publisher: Elsevier Science
Publication Date: 2018-11-02
Edition: 1



Principles of Secure Processor Architecture Design
ISBN: 9781681730011
Authors: Jakub Szefer, Margaret Martonosi
Publisher: Morgan & Claypool Publishers
Publication Date: 2019



The Rootkit Arsenal: Escape and Evasion .
ISBN: 9781449661229
Authors: Bill Blunden
Publisher: Jones & Bartlett Publishers
Publication Date: 2009-05-04



Trusted Computing Platforms
ISBN: 9783319087443
Authors: Graeme Proudler; Liqun Chen; Chris Dalton
Publisher: Springer Nature
Publication Date: 2015-01-08

Brightspace learning management system (LMS)

Access the course via Purdue's Brightspace learning management system. It is strongly suggested that you explore and become familiar not only with the site navigation, but also with the content and resources available for this course. See the Student Services widget in Brightspace for Technology Resources, Academic Resources, Campus Resources, and Health and Well-Being Resources.

Assignments

Table 1: Project milestones by week

Week	Reading	Homework	Project
1	What is HW Security (PDF)	Set up Bootable Cross Compiler	
2	Processor Security Internals (PDF)	Boot QEMU-ARM Bootloader	
3	TPM Specification (PDF)	Set up and Ensuring Secure Boot	Team formation (pre-proposal)
4	Capstone Paper (USENIX '23)		
5	DiskSpy paper (USENIX '25)	TEE on USBArmory (FIDO key)	
6	Posthammer: Pervasive Browser-based Rowhammer Attacks with Postponed Refresh Commands (USENIX '25)		Project Proposal
7	Hot Pixels (USENIX '23)	Cache-Side Keystroke Recovery	
8	CONQUER (NDSS '23)		
9	Warp/Spectre/Downfall		Project Update 1
10	Rowhammer in ARM (ASHES '18)	TrustZone Quote Issuance and Validation	
11	Hidden ARM instr (HASP '20)		
12	ProcessorFuzz (HOST '23)		Project Update 2
13	The Art of Sillicon RE (2014)	Chip Reverse Engineering Tooling	
14	A Security Architect's View of Printed Circuit Board Attacks (USENIX '25)		
15			Final Presentation

WARNING: some of the assignments may change. Please refer to the by-week organization on the content tab to make sure you are up to date.

Grading Scale

Per the Purdue University Academic Regulations, the following grades shall be available to be assigned by the instructors and reported when they are called for by the registrar:

A+

A: Highest passing grade.

A-

B+

B

B-

C+

C

C-

D+

D

D-: Lowest passing grade; marginally passing minimal objectives of the course.

E: Conditional failure; failure to achieve minimum objectives, but only to such limited extent that credit can be obtained by examination or otherwise without repeating the entire course. This grade represents failure in the course unless and until the record is duly changed within one semester. It cannot be improved to a grade higher than D. (See section E.) When an instructor reports a grade of E, they shall file in the departmental office a statement of what is required of the student to receive the passing grade.

F: Failure; failure to achieve minimal objectives of the course.

The student must repeat the course satisfactorily in order to establish credit in it.

Attendance Policy

Class attendance and participation is important for your success in this class. While I will not take attendance, I will note participation and engagement with the class.

Course Schedule

1.3 Lecture Outline

Week(s)	Lecture Topics
PART I	
1	Introduction & Boot Process
2	Basic HW Security Features
3	Secure Co-Processors
4	Advanced CPU security Features
5	Remote Attestation
6	Trust Domains
7	Accelerator Security

Week(s)	Lecture Topics
PART II	
8	SW Side Channels
9	CPU μArch attacks
10	Rowhammer/Aging
11	Firmware Update & Exploits
12	Firmware & HW Fuzzing
13	PUF & HW Assisted Authentication
14	VLSI pipeline attacks
15	Closing thoughts & Semester Review

Academic Integrity

Academic integrity is one of the highest values that Purdue University holds. Individuals are encour-

aged to alert university officials to potential breaches of this value by either emailing integrity@purdue.edu

or by calling 765-494-8778. While information may be submitted anonymously, the more information is

submitted the greater the opportunity for the university to investigate the concern. More details are available

on our course Brightspace under University Policies and Statements.

AI Policy

Overall, please disclose your use of AI. While it is not strictly forbidden to use, I will test your knowledge. If you provide a deliverable with AI without showing *mastery* of the content, you will fail on that assignment. Be prepared to explain your deliverable in an oral exam.

Nondiscrimination Statement

Purdue's full Nondiscrimination Policy Statement is included in the Academic Resources table on your Brightspace homepage.

Accessibility

Every member of our course should be able to access, use, and learn from the materials we share. This includes all course related digital content that you and I share in the course. This approach helps promote equal access for everyone at Purdue and is mandated federally by Title II of the Americans with Disabilities Act (ADA). We will work together to provide this access within our Brightspace course.

- My part, as the instructor, is to make sure all course materials shared to Brightspace, such as documents, slides, videos and audio, and images, meet accessibility guidelines and to assist you in making sure anything you share is accessible. If we can do anything further to support your access to and engagement with course materials, please notify me.
- Your role, as a student, is to make sure anything you post for other students to engage with is also accessible, such as peer grading, peer feedback, and discussion board posts. This expectation is built into all course assignments that require you to post to Brightspace.
- A good starting place for students and instructors is to bookmark and review the Innovative Learning Accessibility Checklist for guidance on creating accessible materials. We need to work together to make sure all materials on our course Brightspace are accessible.
- If you select materials to share on our Brightspace from Purdue Libraries catalog or databases, best practices include choosing items that:
    - Can be downloaded in full
    - Are available in EPUB or HTML formats
    - Include alternative text for written materials or captions for audio/visual content

Accommodations

Purdue University strives to make learning experiences accessible to all participants. If you anticipate or experience physical or academic barriers based on disability, you are encouraged to contact the Disability Resource Center at: drc@purdue.edu or by phone: 765-494-1247, as soon as possible.

If the Disability Resource Center (DRC) has determined reasonable accommodations that you would like to utilize in my class, you must release your Course Accommodation Letter to me. Instructions on sharing your Course Accommodation Letter can be found by visiting: How To Use Your Course Accommodation Letter. Additionally, you are strongly encouraged to contact me as soon as possible to discuss implementation of your accommodation.

Mental Health/Wellness Statement

If you find yourself beginning to feel some stress, anxiety and/or feeling slightly overwhelmed, **try** Therapy Assistance Online (TAO), a web and app-based mental health resource available courtesy of Purdue Counseling and Psychological Services (CAPS). TAO is available to all students at any time by creating an account on the TAO Connect website, or downloading the app from the App Store or Google Play. It offers free, confidential well-being resources through a self-guided program informed by psychotherapy research and strategies that may aid in overcoming anxiety, depression and other concerns. It provides accessible and effective resources including short videos, brief exercises, and self-reflection tools.

If you need support and information about options and resources, please contact or see the Office of the Dean of Students. Call 765-494-1747. Hours of operation are M-F, 8 a.m.-5 p.m.

If you find yourself struggling to find a healthy balance between academics, social life, stress, etc., sign up for free one-on-one virtual or in-person sessions in West Lafayette with a Purdue Wellness Coach at RecWell. Student coaches can help you navigate through barriers and challenges toward your goals throughout the semester. Sign up is free and can be done on BoilerConnect. Students in Indianapolis will find support services curated on the Vice Provost for Student Life website.

If you're struggling and need mental health services: Purdue University is committed to advancing the mental health and well-being of its students. If you or someone you know is feeling overwhelmed, depressed, and/or in need of mental health support, services are available. For help, such individuals should contact Counseling and Psychological Services (CAPS) at 765-494-6995 during and after hours, on weekends and holidays, or by going to the CAPS offices in West Lafayette or Indianapolis.

Emergency Preparedness

In the event of a major campus emergency, course requirements, deadlines and grading percentages are subject to changes that may be necessitated by a revised semester calendar or other circumstances beyond the instructor's control. Relevant changes to this course will be posted onto the course website or can be obtained by contacting the instructors or TAs via email or phone. You are expected to read your @purdue.edu email on a frequent basis.

See Purdue's Information on Emergency Preparation and Planning. This website covers topics such as Severe Weather Guidance, Emergency Plans, and a place to sign up for the Emergency Warning Notification System. I encourage you to download and review the Emergency Procedures Guide.

The first day of class, I will review the **Emergency Preparedness plan for our specific classroom**. Please make note of items like:

- Evacuation areas are In front of MSEE, behind MSEE on Northwestern Ave, or between BHEE and KNOY.

  The atrium of the MSEE building is a secondary location.
- For shelter in place (e.g., tornado): Ground Floor Hallway(s) GH01, GH02, GH03, and GH06 away from exterior entrances/ glass.
- There is no building specific shelter-in-place guidance for threats (e.g., active shooter) in our building

Student-Related Policies

The Vice President for Ethics & Compliance website includes a list of Student Policies. Among those is the Violent Behavior policy, which explains that Purdue is committed to providing a safe and secure campus environment for members of the university community. Purdue strives to create an educational environment for students and a work environment for employees that promote educational and career goals. Violent behavior impedes such goals. Therefore, violent behavior is prohibited in or on any University facility or while participating in any university activity.

Grade Appeals Process

At the end of the course, after final course grades have been assigned, students who believe their grade was assigned contrary to what is outlined in the course syllabus, has been assigned arbitrarily, or has been reduced too severely due to a violation of course academic integrity policies, may submit an appeal request to have this final course grade reviewed. The University's Grade Appeal Process is outlined in the University Catalog and is also referenced on the website for the Office of Student Rights and Responsibilities (OSRR).

Staff in the OSRR office are able to assist in determining if a Grade Appeal is appropriate. Generally, students should not expect a grade appeal to be successful if there is a lack of information to support the appeal or if the student merely thinks that the assigned grade is "unfair". Students must also attempt to resolve the matter informally with the course instructor prior to beginning the appeal process.

Basic Needs Program

If you are facing challenges securing basic needs such as food, housing, transportation, health services, or access to technology or childcare resources and believe this may affect your performance in the course, please contact the Office of the Dean of Students (ODOS) to help coordinate with community resources. These services vary by location. In West Lafayette, see the Basic Needs Program website, or email basicneeds@purdue.edu. To connect with a Student Support Generalist on the Indianapolis campus, contact them by phone at 765-495-7797 or email studentlifeindy@purdue.edu.

Course Evaluation

Toward the end of this semester, you will be provided with an opportunity to give feedback on this course and your instructor. Purdue uses an online course evaluation system, and I will not have access to this anonymous feedback until after final grades are submitted. You will receive an official email from evaluation administrators with a link to the online evaluation site and will receive a prompt to complete the survey when you login to Brightspace. The subject line will be: Please take 2-5 minutes to complete the survey. Check your “Junk E-mail” folder occasionally to be sure the evaluation emails were not accidentally routed there. Your participation is an integral part of this course, and your feedback is vital to improving education at Purdue University. I strongly urge you to participate in the evaluation system.

This syllabus is subject to change. You will be notified of any changes as far in advance as possible via an announcement on Brightspace. Monitor your Purdue email daily for updates.